

Andrea SIGNORETTI

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Profile

Applied mathematics and CS student (X24) at École Polytechnique, specializing in machine learning. I like building agentic LLM systems and training models for applied solutions in high-stakes fields. I want to contribute to Human-Compatible AI research, to keep humans in the loop in secure and explainable settings. Seeking a **6-month final-year research internship (from Summer 2027)** in applied AI.

Education

École Polytechnique (X24) - Cycle Ingénieur Polytechnicien *Palaiseau, FR - 2024-2028*

M.Sc. in Applied Mathematics and Computer Science- 1st Engineering school in France

- **Relevant coursework:** Probability theory for ML, Operational Research, Topological Data Analysis, Optimization for AI, Deep Learning, Reinforcement Learning.
- **Lab Project** (DL for CV) : SSv2 action prediction - trained TSM and video-transformer models from scratch in PyTorch
- **Research Project** (Operational Research - EDF Osiris Lab): Simple power distribution network expansion planning in decomposed networks under supervision of Cécile Rottner and Sandie Balaguer

Lycée Henri IV - Classe Préparatoire (MPSI / MP*) *Paris, FR - 2022-2024*

Two-year intensive preparation for the national competitive entrance exams (mathematics, theoretical CS, physics).

- GPA : 4.0/4.0 (A+) - Admitted to Ecole Polytechnique and ranked 6th at ENS Lyon in the CS track.

Research & Professional Experience

Panthera Solutions - Applied AI Engineer (Intern) (06/2026 - 09/2026) *Monaco, MC*

- Led research in MAS personalization in a high-stakes environment: integrated investor profile metadata in the context to efficiently challenge user's decision with XAI and sourced feedback; validated on real use cases in a proof-of-concept.
- Designed the agentic layer of the six-stage decision workflow (from ideation to monitoring) for the Decision GPS™ application : evidence-based LLM reasoning pipeline to track behavioral-finance criteria and causal reasoning.
- Re-designed the monitoring section : developed a MAS with Research, Process, Challenge agents and a data API access (EODHD) to monitor evidence-based decisions.

Orange Innovation - Computer Science Research Project (06/2025 - 06/2026) *Paris, FR*

- Designed a three-layer explainable multi-agent system (JaCaMo) for network anomaly detection and RCA over RDF/OWL knowledge graphs, combining 54 deterministic SPARQL agents with 12 correlation diagnosers and an LLM explainer
- Built a reproducible 27-graph benchmark showing 0.815 F1 vs 0.508 (standalone LLM) and 100% robustness under noise.
- Co-authoring a research paper on multi-agent synergistic reasoning, targeting JFSMA @ PFIA 2027.

French Army Intelligence - Full-Stack Engineer Intern (10/2024 - 04/2025) *Strasbourg, FR*

- Developed a secure workspace application for OSINT data-collection (PHP, SQL, TypeScript), now used by 6 intelligence agencies in France.
- Led the user-oriented design from need survey to delivery with high autonomy. Received an official Commendation Letter.

Awards & Honors

- **1st Prize** (2025) - BCGX DataQuest Hackathon (AgriCredit, agricultural investment prediction).
- **Official Commendation Letter** (2025) - French Ministry of Defense, for my commitment and ingenuity during my internship in intelligence services.
- **2nd Prize** - National Mathematics Olympiads for 11th grade students (2021).

Personal Projects - *portfolio*

ScholarMates : an agentic HITL workspace for writing research papers end to end - RAG over bibliography, OCR, ArXiv / Google Scholar search, and role-specialized agents (organizer, writer, reviewer).

EarDrop : learns to recognize the 30 most-spoken languages by ear, comparing a from-scratch acoustic classifier against a fine-tuned Whisper-small, and shows the audio cues that separate one language from another.

AgriCredit (BCGX DataQuest Hackathon, 1st prize): predicted the production outcome of agricultural investments in Morocco from hydrographic and economic data to guide where capital should go.

Extracurriculars

XProjets (École Polytechnique Junior Enterprise) - Vice President. Leading a 45-person team on a 400k€ data project for a leading European AI lab, overseeing quality and volume control. +180% in revenue from 2025 to reach 550k€.

X-Forum - Treasurer & Head of IT. Responsible for the accounting and billing of the 1.3M€ career fair, gathering over 200 companies and 1500 students. Led the full redesign and development of the registration platform and [public website](#).

Varsity Basketball - Captain. Twelve competitive seasons leading the team.

Technical Skills

ML / DL	Python, PyTorch, multi-agent systems, RAG, LoRA, knowledge graphs (RDF / OWL / SPARQL)
Engineering	TypeScript, FastAPI, PHP, SQL, Flask, React / Next.js, PostgreSQL, Docker, Git, CI/CD, Linux
Spoken	French (native), Italian (native), English (fluent, C1-C2)